

Emil Schmitz

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New York, NY (US permanent resident)

Work Experience

AI Engineer, *Web Computing (Sparkassen Finance Group)* – Germany (remote) Mar 2025 – Apr 2026

- Reduced manual workflows from minutes to seconds for a 280k+ user ticketing system by engineering semantic search and AI ticket routing using PyTorch, Hugging Face, and Elasticsearch.
- Automated thousands of hours of manual compliance work annually by engineering an LLM pipeline for flagging security incidents. Created Helm chart for deployment on Kubernetes cluster.
- Developed *AI direct answers* for internal search engine using retrieval-augmented generation (RAG).
- Worked closely with technical and non-technical stakeholders to build perfectly tailored custom solutions.
- Secured company multi-million Euro project by building a prototype for an agentic credit-rating system.

Math Teacher, *SchuleEins* – Berlin, Germany Dec 2023 – Jan 2025

- Taught math to students aged 14-18.

Math and CS Tutor, *Freelance* – Berlin, Germany Aug 2023 – Feb 2025

- Tutored college-level mathematics and computer science.

Quantitative Developer, *Goldmarie Finanzen* – Berlin, Germany Jul 2022 – Jan 2023

- Engineered and backtested algorithmic trading strategies using NumPy, Pandas and SciPy, ensuring production readiness via statistical validation and Azure Cosmos DB integration.
- Researched and implemented risk-management protocols to automate compliance with financial regulations.

Projects

Prosocial Ranking Challenge 2024

- Won \$11,000 in UC Berkeley's *Prosocial Ranking Challenge* by developing a novel AI-based social-media ranking system.
- Built a backend system using Celery and Redis for asynchronous post scoring via custom-trained Hugging Face sentence transformers, deployed on AWS.

Technical Skills

Languages: Python, JS/TS (React), Rust, C++

ML & AI: LiteLLM, Langchain, LlamaIndex, MCP, PyTorch, Hugging Face, Scikit-learn, W&B, DVC

Backend: FastAPI, Pydantic, Celery, Redis, Elasticsearch, MongoDB, Nginx

Tools: Linux, Git, Agentic Coding Tools, Docker, Kubernetes, Prometheus/Grafana, Jenkins CI/CD, AWS

Education

MS Mathematics – Technische Universität Berlin 2022

- GPA: 3.8 (US-equivalent)
- Specialized in applied probability and statistics.

BS Mathematics – Universität Heidelberg 2020